LLM-Based Report Generation

SEMINAR PRESENTATION

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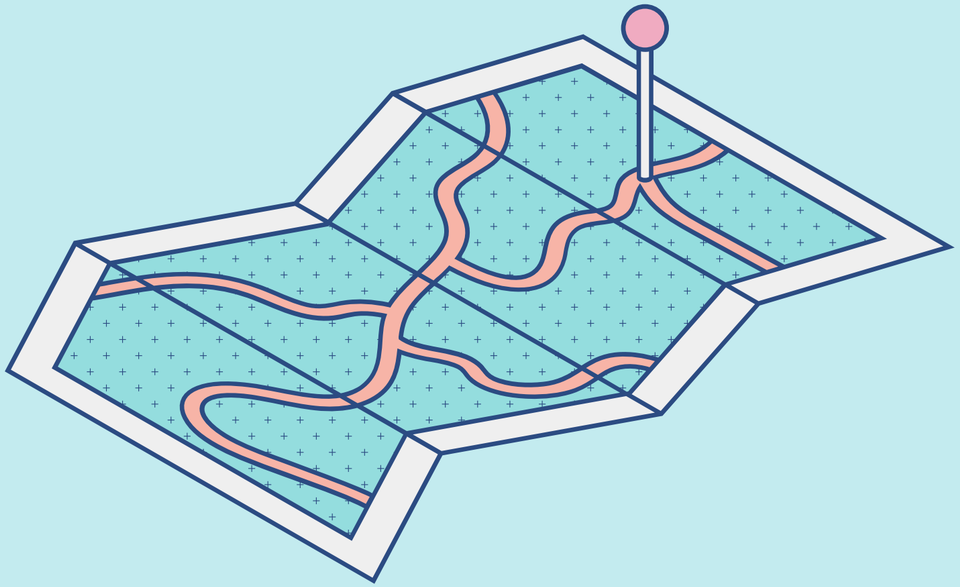
I. Introduction

Overall

This project implements an LLM-based report summarization system that transforms structured JSON system reports into clear natural language summaries for dashboards and system monitoring, enabling fast and efficient interpretation of operational data.

II. DATASET

The dataset is a programmatically generated daily transaction report that simulates a tuition payment system, containing aggregated statistics and detailed transaction records in structured JSON format for LLM-based summarization, while ensuring reproducibility and data privacy.

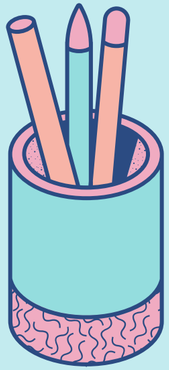


III. API LIST

GenerateDailyReport

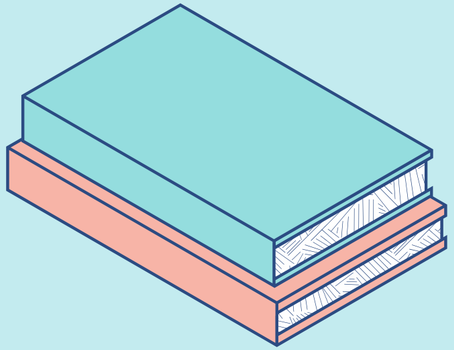
- Input: None
- Output: JSON daily transaction report
- Description: Generates a daily transaction report on demand by simulating tuition payment activities. The report includes aggregated statistics such as total transactions, successful and failed payments, total processed amount, and a list of individual

III. API LIST



ReadExistingReport

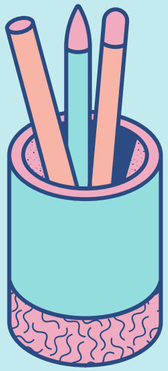
- Input: None
- Output: JSON report loaded from file
- Description: Reads and returns an existing system report stored in dataset.json. This service allows the system to retrieve previously generated reports for further processing or analysis.



III. API LIST

HealthCheck

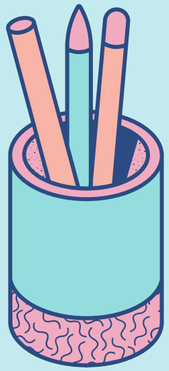
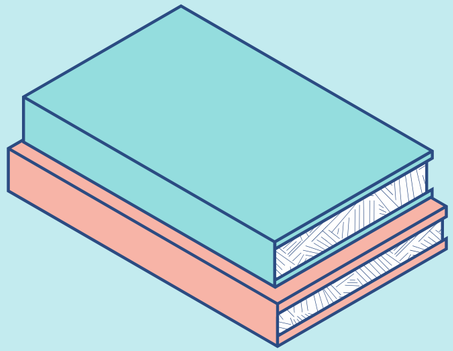
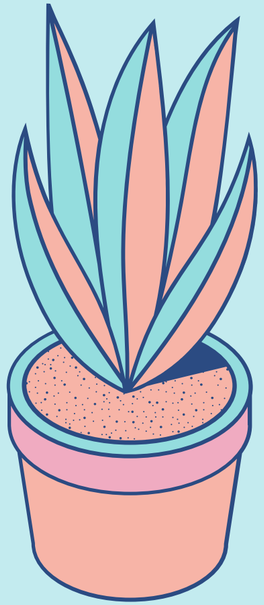
- Input: None
- Output: service status message
- Description: Revokes current access token (logout).



III. API LIST

SummarizeReport

- Input: mode (query parameter) (mode could be concise or detailed)
- Output: Natural language summary of the system report
- Description: Loads a structured transaction report from dataset.json and generates a human-readable summary using a Large Language Model (Gemini). The summarization behavior is controlled by the mode parameter, allowing the system to produce either a concise dashboard-style summary or a detailed operational report.



THANK YOU FOR LISTENING